SUPPORTING ADULT GUIDANCE

Order of operations and calculator judgement

Number Systems and Calculator Judgement · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson revisits the order of operations at a higher level — with powers, roots, negative numbers and fraction bars — and connects it to calculator use. Learners choose values and see the order of working, then use an on-screen calculator that draws what it actually heard beside what is written, and fix the keys with brackets. They judge calculator displays by estimating, and hit target values by placing brackets.

Driving Question: *When does the calculator get it wrong — and when do I?*

The mathematics, in plain terms

Calculators follow a fixed precedence: brackets; powers and roots; × and ÷ left to right; + and – left to right. A leading minus sign is applied after a power, so $-3^2 = -9$ — the convention used by mathematicians and almost all calculators. In written mathematics, a fraction bar and a radical (√) sign act as grouping symbols, so when a calculation is entered on one line, those groups must be bracketed. The lesson's calculator applies √ to the number or bracket straight after it; some real calculators open a bracket automatically after √, or have a fraction template. Whatever the model, the habit is the same: check that the display shows what was meant.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — What the calculator heard

Work out the fraction with numerator 6 + 9 and denominator 2 + 3.

1. Keyed as it looks: 6 + 9 ÷ 2 + 3	the calculator works out 9 ÷ 2 first
2. It heard 6 + 4.5 + 3.	= 13.5
3. The fraction bar groups each part, so key brackets: (6 + 9) ÷ (2 + 3)	15 ÷ 5
4. Estimate check: 15 ÷ 5 is small, not 13.5.	= 3

The answer is 3. Each hidden group needed its own brackets.

Example 2 — A minus sign and a square

Compare -4^2 with $(-4)^2$.

1. -4^2: the square belongs to the 4 only. Square first.	$4^2 = 16$
2. Then apply the minus sign.	$-4^2 = -16$
3. $(-4)^2$: the brackets make -4 the number being squared.	-4×-4
4. A negative times a negative is positive.	$(-4)^2 = 16$

$-4^2 = -16$ but $(-4)^2 = 16$. To square a negative number, key it in brackets.

See it before fixing it

Each calculation in the main activity is preloaded with the keys someone would naturally type. Learners press = first and see what the calculator heard, drawn as written mathematics, and the order it worked in. Only then do they repair the keys. The rule 'bracket the hidden groups' comes from seeing the mismatch, not from being told. The estimation questions then turn this into a habit: estimate, then trust.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	First things first: choose values for calculations with powers, roots and fraction bars, and see the order of working.	~7 min
Movement	What the calculator heard: press =, compare with what is written, and fix the keys with brackets.	~15 min
Reflection	Can I trust it? Estimate, then decide whether a display is sensible or a slip.	~8 min
Creativity	Bracket puzzles: the same keys, different brackets, different targets.	~10 min
Rest	The order of operations as a shared agreement.	~2 min
Nutrition	Intellectual: trust in your own judgement over a machine's answer.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Keying a fraction as it looks, without brackets (6 + 9 ÷ 2 + 3).
Ask: “What does the fraction bar divide by?”

Believing $-4^2 = 16$.
Ask: “Which number does the square belong to?” Then compare with $(-4)^2$.

Working × before ÷ (or + before –) instead of left to right: 24 ÷ 4 × 2 as 3.
Ask: “Which comes first, reading from the left?”

Taking the root of only the first number under the √ bar.
Ask: “How far does the √ bar reach?”

Trusting any number the calculator shows.
Ask for a rough estimate first, then compare.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the order of operations, squares and square roots. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner gets 810 for 360 divided by 4 × 9 and accepts it. The aim is a question that lets them find the slip themselves.

- “Roughly, what is 4 × 9?”
- “About how many 36s are there in 360?”

- “Does 810 fit with that?”
- “What did the calculator do first with the keys you pressed?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners use calculators in this lesson?

Yes — deliberately. The lesson is about using them well. Learners also work some answers out by hand to check them.

My learner's calculator behaves differently.

Calculators vary: some open a bracket after √, some have a fraction template. Ask the learner to check what the display shows. The habit matters more than the model.

Is $-3^2 = -9$ just a calculator rule?

No. It is the convention used in mathematics generally: a power applies only to what is directly before it.

What comes next?

Lesson 4: negative and fractional indices.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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